

SMC Estimators for KL Divergence

Ari Blondal, Skylar Gu, Rahma Nouaji

Abstract

Accurate estimation of Kullback-Leibler (KL) divergence between large language models is essential for monitoring distributional shifts during fine-tuning, detecting model collapse, and ensuring safe deployment. However, direct computation is intractable, and standard Monte Carlo estimators exhibit prohibitively high variance due to the long-tailed nature of language model distributions. While importance sampling offers a potential solution, the design of effective proposal distributions remains an open challenge.

This paper presents preliminary investigation into many importance sampling strategies for KL divergence estimation. Experiments are run both on Zipfian distributions modeling real language models and on smaller open-weight models, such as Qwen2.5-1.5B and Qwen2.5-0.5B.

Our analysis is inconclusive, pointing towards the need for more experiments, potentially with greater computational resources.

Code: https://github.com/ablondal/KL_estimation

1 Introduction

The Kullback-Leibler divergence (Kullback & Leibler, 1951) is a commonly used metric for measuring the statistical divergence between two probability distributions. In the context of language models, and more specifically LLMs, this divergence is used during training to show convergence or for regularization purposes.

For two distributions P and Q over the same domain $\mathcal{X}$, the KL divergence $\text{KL}(P\|Q)$ is defined as

$$\text{KL}(P\|Q) := \sum_{\mathbf{x} \in \mathcal{X}} P(\mathbf{x}) \log \frac{P(\mathbf{x})}{Q(\mathbf{x})} = \mathbb{E}_{\mathbf{x} \sim P} \left[\log \frac{P(\mathbf{x})}{Q(\mathbf{x})} \right].$$

In many cases, such as when the domain $\mathcal{X}$ is enumerable or the distributions are Gaussian, this divergence is calculable. On the other hand, it is a difficult metric to calculate when working with autoregressive language models. Given a finite alphabet Σ , the domain $\mathcal{X}$ consists of all finite EOF-terminated strings Σ_{EOF}^* . Note that $\text{EOF} \notin \Sigma$.

In particular, Σ_{EOF}^* is infinite, so exact divergence calculation would have to use the transformer-based language encoder architecture of state-of-the-art language models. Unfortunately, this architecture exhibits chaotic behaviour (Inoue et al., 2022), indicating the intractability of dealing with models directly.

Instead, the best-known methods for estimating KL divergence between autoregressive language models uses Monte Carlo sampling, sampling strings $\mathbf{x}$ directly from P . That is, given M independent samples $\mathbf{x}^{(i)} \sim P^M$ drawn from the language model P , the KL divergence is estimated as follows:

$$\hat{\mu}_{\text{MC}} = \frac{1}{M} \sum_{i=1}^M \log \frac{P(\mathbf{x}^{(i)})}{Q(\mathbf{x}^{(i)})}.$$

To increase the accuracy of our estimation, we can increase M , as the variance of $\hat{\mu}$ scales proportionally to $1/\sqrt{M}$. Alternatively, variance can be reduced by sampling from a better distribution. Sampling directly from P uses no information about Q , so the goal of the

project was to test various tractable distributions from which to sample from, that would result in faster convergence of $\hat{\mu}$.

Given a distribution r on $\mathcal{X}$, where $r(x) = 0$ only if $P(x) = 0$, we can calculate the KL divergence by sampling from r instead of P .

$$\begin{aligned}\text{KL}(P\|Q) &= \mathbb{E}_{\mathbf{x} \sim P} \left[\log \frac{P(\mathbf{x})}{Q(\mathbf{x})} \right] = \mathbb{E}_{\mathbf{x} \sim r} \left[\frac{P(\mathbf{x})}{r(\mathbf{x})} \log \frac{P(\mathbf{x})}{Q(\mathbf{x})} \right] \\ \hat{\mu}_{\text{IS}} &= \frac{1}{M} \sum_{i=1}^M \frac{P(\mathbf{x}^{(i)})}{r(\mathbf{x}^{(i)})} \log \frac{P(\mathbf{x}^{(i)})}{Q(\mathbf{x}^{(i)})}.\end{aligned}$$

Since M can simply be increased to increase the accuracy of any estimation, it is important that any proposal distribution r is computationally tractable. Therefore, like P and Q , all the distributions we propose are autoregressive. That is to say, the probability measure of a string $\mathbf{x} = (\mathbf{x}_j)_{j=1}^N$ of length N can be decomposed into token-wise probabilities.

$$r(\mathbf{x}) = \prod_{j=1}^N r(\mathbf{x}_j \mid \mathbf{x}_{<j}).$$

In order to evaluate our results, we have to run them against a known baseline. Recently, Amini, Vieira and Cotterell introduced a Rao-Blackwellized version of this calculation, leveraging the extra information available at each inference step to reduce the variance of $\hat{\mu}$ and increase the rate of convergence of the estimation (Amini et al., 2025).

In this paper, we do not employ their method, instead using it as a baseline against which to compare our proposal distributions. It can, however, easily be adapted to any of our autoregressive proposal distributions, so we focus on finding a distribution whose KL estimation has empirically lower variance than p .

The remainder of this paper is organized as follows: we provide background on importance sampling (Section 2), present our proposal distributions (Section 3), detail our experimental setup and present our results on synthetic distributions and language models (Section 4).

2 Background

Recall the standard approach to estimating KL divergence, the naive Monte Carlo (MC) method: sample M independent sequences $(\mathbf{x}^{(i)})_{i=1}^M \sim P^M$ and compute:

$$\hat{\mu}_{\text{MC}} = \frac{1}{M} \sum_{i=1}^M \log \frac{P(\mathbf{x}^{(i)})}{Q(\mathbf{x}^{(i)})}.$$

This estimator is unbiased: $\mathbb{E}[\hat{\mu}_{\text{MC}}] = \text{KL}(P\|Q)$. Its variance scales as $\theta(1/\sqrt{M})$, meaning accuracy improvements require quadratic numbers of samples.

Limitation: Language model distributions are heavily long-tailed, with most probability mass concentrated on a small subset of tokens. This leads to high variance in the MC estimator, as rare high-divergence tokens are undersampled.

2.1 Importance Sampling

Importance sampling (IS) offers potential variance reduction by sampling $\mathbf{x}^{(i)} \sim r$ from an alternative proposal distribution r instead of P :

$$\hat{\mu}_{\text{IS}} := \frac{1}{M} \sum_{i=1}^M w_i f_i, \quad w_i := w_i(\mathbf{x}^{(i)}) = \frac{P(\mathbf{x}^{(i)})}{r(\mathbf{x}^{(i)})}, \quad f_i := f_i(\mathbf{x}^{(i)}) = \log \frac{P(\mathbf{x}^{(i)})}{Q(\mathbf{x}^{(i)})}.$$

We refer to w_i as importance weights.

Requirements for valid proposals:

1. **Support condition:** $r(x) > 0$ whenever $P(x) > 0$ (absolute continuity).
2. **Tractability:** The density $r(x)$ must be computable and efficient to sample from.

2.2 Self-Normalized Importance Sampling

2.2.1 Estimator Definition

In practice, importance sampling can still have very high variance. This can be the case when the divergence between r and P is quite high, leading to high variance in the weights. Self-normalized importance sampling (SNIS) addresses this by normalizing the estimation with respect to the calculated weights, and can lead to more stable estimation of the KL divergence (Kallus & Uehara, 2019).

$$\hat{\mu}_{\text{SNIS}} := \frac{\sum_{i=1}^M w_i f_i}{\sum_{i=1}^M w_i} = \sum_{i=1}^M w'_i f_i \quad \text{where} \quad w'_i = \frac{w_i}{\sum_{i=1}^M w_i}.$$

SNIS exchanges variance for bias, and poor proposal distributions r lead to poor estimations, albeit ones that converge more quickly. For our purposes, this means that detected bias (estimation error when compared against a known baseline) serves as a more reliable metric of the proposal distribution's performance.

The performance of SNIS depends critically on the proposal distribution $r(x)$: poor proposals lead to high weight variance, low effective sample size (ESS), and degraded estimation quality. The goal of this work is to identify effective proposal distributions for KL divergence estimation between language models.

In this paper, we primarily use SNIS, not only because we do not require the normalizing constant of $r(x)$, but also because it leads to more stable estimation of the KL divergence (Kallus & Uehara, 2019).

2.2.2 Bias, Asymptotic Properties, and Metrics

SNIS has $\mathcal{O}(1/M)$ finite-sample bias (see Appendix for full derivation):

$$\mathbb{E}[\hat{\mu}_{\text{SNIS}}] = \mu + \frac{1}{M} \left[\frac{\mathbb{E}[w^2 f]}{\mathbb{E}[w]^2} - \frac{\mathbb{E}[w f] \mathbb{E}[w^2]}{\mathbb{E}[w]^3} \right] + \mathcal{O}(1/M^2)$$

where $\mu = \mathbb{E}_P[f(\mathbf{x})]$ is the true KL divergence. As $M \rightarrow \infty$, SNIS is strongly consistent. The estimation $\hat{\mu}_{\text{SNIS}}$ converges to μ , and the error is asymptotically normal (Agapiou et al., 2017; Chopin et al., 2020):

$$\begin{aligned} \sqrt{M}(\hat{\mu}_{\text{SNIS}} - \mu) &\rightarrow \mathcal{N}(0, \sigma_{\text{SNIS}}^2) \\ \text{where } \sigma_{\text{SNIS}}^2 &= \frac{\mathbb{E}_r[w^2(f - \mu)^2]}{\mathbb{E}_r[w]^2}. \end{aligned}$$

Evaluating the performance of SNIS is difficult, as it is the ratio of two random variables. As such, we use this asymptotic behaviour to **estimate** the variance of $\hat{\mu}_{\text{SNIS}}$.

$$\begin{aligned} \lim_{M \rightarrow \infty} \text{Var} \left[\sqrt{M} \hat{\mu}_{\text{SNIS}} \right] &= \sigma_{\text{SNIS}}^2 \\ \text{Var}[\hat{\mu}_{\text{SNIS}}] &\rightarrow \frac{1}{M} \frac{\mathbb{E}_r[w^2(f - \mu)^2]}{\mathbb{E}_r[w]^2} \\ \widehat{\text{Var}}[\hat{\mu}_{\text{SNIS}}] &= \frac{1}{M} \frac{\frac{1}{M} \sum_i w_i^2 (f_i - \hat{\mu}_{\text{SNIS}})^2}{\frac{1}{M^2} (\sum_i w_i)^2} \\ &= \sum_{i=1}^M w_i'^2 (f_i - \hat{\mu}_{\text{SNIS}})^2. \end{aligned}$$

In order to evaluate the performance of a particular method, we also calculate the weighted squared error $\widehat{\text{WSE}}$, and the effective sample size $\widehat{\text{ESS}}$ of a sample set:

$$\widehat{\text{WSE}} := \sum_{i=1}^M w'_i (f_i - \hat{\mu}_{\text{SNIS}})^2$$

$$\widehat{\text{ESS}} := \frac{(\sum_{i=1}^M w_i)^2}{\sum_{i=1}^M w_i^2} = \frac{1}{\sum_{i=1}^M (w'_i)^2}.$$

While the variance and mean squared error are measures of the accuracy of the estimation, the effective sample size measures proposal efficiency by quantifying how many independent samples the weighted sample is equivalent to. High ESS ($\approx n$) indicates nearly uniform weights and an efficient proposal, while low ESS ($\ll n$) indicates weight concentration on few samples and a poor proposal. For SNIS, both bias and variance scale inversely with ESS: $|\text{Bias}| \propto \text{ESS}^{-1}$ and $\text{Var}[\hat{\mu}_{\text{SNIS}}] \propto \text{ESS}^{-1}$ (Agapiou et al., 2017). We report ESS alongside error and variance metrics to evaluate proposal quality.

3 Proposal Distributions

3.1 Design Principles

Effective proposal distributions for KL divergence estimation must balance three competing objectives:

1. **Support Coverage:** The proposal must satisfy the absolute continuity condition: $r(x) > 0$ whenever $P(x) > 0$. This ensures all tokens with non-zero probability under P can be sampled, preventing infinite importance weights.
2. **Divergence Targeting:** The optimal proposal $r^*(x) \propto P(x) |\log(P(x)/Q(x))|$ concentrates sampling on high-divergence regions where P and Q differ most. Since this is intractable (it requires knowing the KL integrand we’re estimating), practical proposals must approximate this behavior using tractable surrogates.
3. **Computational Tractability:** For autoregressive language models, proposals must decompose sequentially: $r(x) = \prod_{t=1}^T r(x_t | x_{<t})$, where each conditional distribution is efficiently computable and sampleable.

The Tradeoff: Proposals that strongly emphasize high-divergence regions (targeting objective 2) risk weight concentration and poor ESS, while proposals that broadly cover P ’s support (objective 1) may undersample critical regions. The proposals below explore different points in this tradeoff space.

3.2 Proposal Families

In the following distributions, r , P , and Q refer to $r(x_t | x_{<t})$, $P(x_t | x_{<t})$, and $Q(x_t | x_{<t})$. $\log(P/Q)$ refers to $\log(P(x_{\leq t})/Q(x_{\leq t}))$. This is in order to boost the preexisting bias that a partial sample may have towards one distribution over another, as otherwise tokens on which $P(x) > Q(x)$ and tokens where $Q(x) > P(x)$ may be likely to be chained together, with an overall low log difference.

- **Fixed mixtures:** $r = \lambda P + (1 - \lambda)Q$. We try various $\lambda \in \{0.3, 0.5, 0.7, 0.9\}$. Simple baseline requiring no divergence information.
- **Alpha-mixtures:** $r \propto \alpha P |\log(P/Q)| + (1 - \alpha)P$ for $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$. Explicitly weights high-divergence regions while maintaining coverage.
- **Adaptive log-mixture:** $r \propto P(1 + |\log(P/Q)|)$. First-order Taylor approximation of exponential tilting.
- **Geometric mixtures:** $r \propto P^\beta Q^{1-\beta}$ for $\beta \in \{0.3, 0.5, 0.7, 0.9\}$. Smoothly interpolates between P and Q .

- **Balanced proposals:** (1) $r \propto \sqrt{P \cdot |\log(P/Q)| \cdot Q/(P+Q)}$ (normalized), (2) $r \propto \sqrt{PQ} \cdot |\log(P/Q)|$ (simple), (3) $r \propto \sqrt{PQ \cdot (\log^2(P/Q) + 1)}$ (quadratic). Combine geometric mean with divergence weighting.
- **Exponential family:** $r \propto P \exp(\beta |\log(P/Q)|)$ for $\beta \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$. Exponentially tilts for aggressive targeting.
- **Cross-entropy tilted:** $r \propto P \exp(\beta H_x(P, Q))$ where $H_x = -P \log Q$, for $\beta \in \{0.3, 0.5, 0.7, 0.9\}$. Targets high negative log-probability under Q .
- **Chi-square aware:** $r \propto P \cdot (\frac{P}{Q})^\alpha$ for $\alpha = 0.5$. Optimal for χ^2 divergence. (Only for synthetic experiments.)

For numerical stability of these proposals when computing log-ratios, we apply clipping: $\tilde{L}(x) = \text{clip}(\log(P(x)/Q(x)), -M, M)$ with $M = 50$. This prevents overflow/underflow when $P(x) \gg Q(x)$ or $P(x) \ll Q(x)$. The introduced bias is bounded by: $|\text{KL}(P\|Q) - \widetilde{\text{KL}}(P\|Q)| \leq \sum_{x: |L(x)| > M} P(x) |L(x)|$. For well-behaved distributions, this bias is negligible. We also clip probabilities to avoid division by zero: $P_{\text{safe}}(x) = \max(P(x), \epsilon)$, $Q_{\text{safe}}(x) = \max(Q(x), \epsilon)$ with $\epsilon = 10^{-12}$.

3.3 Comparison Baselines

We evaluate our importance sampling methods against multiple baselines. All experiments include naive Monte Carlo as the fundamental baseline. For synthetic distributions, we additionally compare against the theoretically optimal proposal and measure error relative to exact KL divergence. For full LLM experiments, however, we compare against the state-of-the-art method, a Rao-Blackwellized Monte Carlo method (Amini et al., 2025).

3.3.1 Naive Monte Carlo

The simplest approach samples complete sequences from P and computes:

$$\widehat{\text{KL}}_{\text{MC}} = \frac{1}{M} \sum_{i=1}^M \sum_{t=1}^T \log \frac{P(x_t^{(i)} | x_{<t}^{(i)})}{Q(x_t^{(i)} | x_{<t}^{(i)})}, \quad x^{(i)} \sim P. \quad (1)$$

This estimator is unbiased but typically exhibits high variance, especially when P and Q differ significantly in low-probability regions.

3.3.2 Theoretically Optimal (Intractable) Proposal

The variance-minimizing proposal for KL estimation is:

$$r_t^{\text{opt}}(x_t | x_{<t}) \propto P(x_t | x_{<t}) \cdot \left| \log \frac{P(x_t | x_{<t})}{Q(x_t | x_{<t})} + \text{KL}(P \| Q | x_{\leq t}) \right|, \quad (2)$$

where $\text{KL}(P \| Q | x_{\leq t})$ is the *future* KL conditioned on the current history. This is intractable as it requires knowledge of future divergence.

3.3.3 Rao-Blackwellized Monte Carlo

For tests where calculating the exact KL divergence is intractable, we instead use the following estimation as a baseline.

$$\widehat{\text{KL}}_{\text{RB}} = \frac{1}{M} \sum_{i=1}^M \sum_{t=1}^T \text{KL} \left(P(\cdot | \mathbf{x}_{<t}^{(i)}) \| Q(\cdot | \mathbf{x}_{<t}^{(i)}) \right), \quad \mathbf{x}^{(i)} \sim P. \quad (3)$$

This *partially* Rao-Blackwellized approach eliminates variance from the current token sampling but retains variance from history sampling. It serves as a performance ceiling for our experiments.

4 Experimental Setup & Results

4.1 Evaluation Metrics

Recall the following evaluation metrics from Section 2. Let $w'_i = w_i / (\sum_i w_i)$ be the normalized importance weights. We evaluate proposals using:

- **Baseline KL:** Our baseline KL_0 is either the exact value or the Rao-Blackwellized estimation.
- **Absolute Error:** $|\hat{KL} - KL_0|$
- **Variance:** Empirical SNIS variance $\sum w'_i{}^2 (f_i - \hat{\mu})^2$ (lower is better)
- **WSE:** Empirical weighted squared error $\sum w'_i (f_i - \hat{\mu})^2$ (lower is better)
- **ESS:** Effective Sample Size $1 / \sum_i w'_i{}^2$ (higher is better)
- **Score:** Combined metric $0.7 \times \text{Error} / KL_0 + 0.3 \times \text{Var} / \sigma_{MC}^2$ (lower is better)

4.2 Experiment 1: Synthetic Distribution

We construct synthetic long-tailed distributions that mirror LM statistics, enabling controlled evaluation of importance sampling proposals with exact computation.

Distribution Families. We evaluate proposals across three synthetic distribution families:

- **Zipfian:** $P(k) \propto k^{-\alpha}$ where $\alpha \in [1.0, 1.5]$ controls tail heaviness, matching empirical observations from language model token distributions.
- **Zipfian with Exponential Cutoff:** Combines power-law head with exponential decay in the tail:

$$P(k) \propto \begin{cases} k^{-\alpha} & k \leq k_{\text{cutoff}} \\ k^{-\alpha} \exp\left(-\frac{k-k_{\text{cutoff}}}{\lambda}\right) & k > k_{\text{cutoff}} \end{cases}$$

This captures the empirical observation that LLM distributions exhibit power-law behavior for frequent tokens but faster-than-power-law decay in the long tail.

- **Mixture Distribution:** $P(k) \propto \begin{cases} e^{-0.05k}, & 1 \leq k \leq 100 \\ k^{-1.3}, & 101 \leq k \leq V/2 \\ k^{-0.7}, & V/2 < k \leq V \end{cases}$

with lognormal perturbations.

Creating Model Pairs (P, Q). We generate P from each base distribution family, then create Q by randomly redistributing probability mass on 100 tokens (excluding the top 100). The shift ratio $\rho \in \{0.1, 0.3, 0.5\}$ controls divergence magnitude (low, medium, high).

4.2.1 Discussion

Table 1 reveals that no single proposal distribution achieves uniformly superior performance across all distribution families. Chi-square ($\alpha = 0.3$) emerges as the most robust method, achieving the best combined score across all three families (0.0060, 0.0041, 0.0134) and consistently delivering the lowest absolute error. This robustness stems from its adaptive probability mass allocation, making it effective across varying tail behaviors. Alpha-mixture performance is distribution-dependent. Different α values excel under different conditions: $\alpha = 0.9$ performs best on smooth Zipfian distributions (ESS: 86.44%), while $\alpha = 0.7$ handles more structured distributions effectively. This sensitivity suggests alpha-mixture requires careful tuning for each distribution family.

The Optimal proposal shows surprising limitations, achieving the best score only for Zipfian+Cutoff (0.0027) and ranking third or fourth elsewhere. This counterintuitive result

Distribution	Method	True KL	Abs Error	$\widehat{\text{Var}}[\hat{\mu}]$	ESS (%)	Score
Zipfian $P(k) \propto k^{-1.5}$	Chi-square ($\alpha = 0.3$)	0.3084	0.0003*	0.0139	75.12	0.0060*
	Alpha-mixture ($\alpha = 0.9$)		0.0036	0.0056*	86.44	0.0104
	Alpha-mixture ($\alpha = 0.7$)		0.0031	0.0221	92.60	0.0156
	Optimal		0.0063	0.0092	81.66	0.0179
	Adaptive log-mixture		0.0067	0.0417	96.14*	0.0314
	Naive MC		0.0462	0.0070	100.00	0.1068
Zipfian + Cutoff	Optimal	0.2858	0.0008*	0.0011*	90.81	0.0027*
	Chi-square ($\alpha = 0.3$)		0.0005	0.0041	83.60	0.0041
	Alpha-mixture ($\alpha = 0.9$)		0.0023	0.0008	93.08	0.0062
	Alpha-mixture ($\alpha = 0.5$)		0.0030	0.0203	97.96*	0.0217
	Alpha-mixture ($\alpha = 0.7$)		0.0089	0.0066	96.49	0.0264
	Naive MC		0.0247	0.0000	100.00	0.0605
Mixture Heavy head + power middle + flat tail	Chi-square ($\alpha = 0.3$)	0.4909	0.0055*	0.0787	30.27	0.0134*
	Alpha-mixture ($\alpha = 0.7$)		0.0073	0.0922	72.26	0.0169
	Optimal		0.0156	0.0181*	48.19	0.0235
	Alpha-mixture ($\alpha = 0.9$)		0.0156	0.0347	58.04	0.0247
	Alpha-mixture ($\alpha = 0.5$)		0.0257	0.2613	84.06*	0.0552
	Naive MC		0.0434	0.2354	100.00	0.1053

Table 1: Top 5 proposal distributions by combined score across three synthetic distribution families. All experiments use $n = 300$ samples, vocab size = 3000, medium divergence ($\rho = 0.3$). Score combines accuracy (70%) and variance (30%).* indicates best performance in each metric.

likely reflects finite-sample estimation errors in constructing the proposal, which introduce bias that dominates the combined score for complex distributions. All importance sampling methods substantially outperform Naive MC, which consistently achieves the worst scores (0.1068, 0.0605, 0.1053) despite maintaining 100% ESS. The best methods reduce absolute error by 18-24 $\times$ while achieving 51-94% variance reduction, validating the value of sophisticated proposal distributions for KL estimation.

4.3 Experiment 2: Qwen2.5

Further experiments were run on two different sizes of Qwen2.5 models.

$$P = \text{Qwen2.5-1.5B}$$

$$Q = \text{Qwen2.5-0.5B}$$

These were designed as exploratory experiments on as many different proposal distributions as possible. For this reason, and due to computational constraints, the following parameters were chosen:

1. The temperature of each model was set to 0.4.
2. Samples were cut off after 20 generated tokens.
3. 1000 samples were taken for each distribution.
4. Each model was prompted with the phrase “An interesting fact is that”.

Even so, not all proposed distributions were tested.

4.3.1 Discussion

The models were run with 16-bit floating point precision. This is why the effective sample size of Naïve MC is not listed as 100%. It is likely that the conversion between 16- and 64-bit precision caused other rounding errors, but overall, the results in Table 2 are clear.

We chose to use self-normalized importance sampling as it provided estimations closer to that of the baseline, **across the board**. Still, our estimations were very bad in many cases,

Baseline	KL estimate	$\widehat{WSE}$	$\widehat{Var}[\hat{\mu}]$	ESS (%)	Score
Rao-Blackwellized MC	12.8165	60.3115	0.0603	100.00	0.2043
Naive MC	12.9351	86.7993	0.0886	98.45	0.3065
err: 0.1186					
Method	Absolute Error	$\widehat{WSE}$	$\widehat{Var}[\hat{\mu}_{SNIS}]$	ESS (%)	Score
Alpha-mixture ($\alpha = 0.9$)	1.4984	66.94	1.2976	2.86	4.4771
Alpha-mixture ($\alpha = 0.7$)	0.7126	75.42	0.3656	12.47	1.2774
Alpha-mixture ($\alpha = 0.5$)	0.4938	86.30	0.3529	27.82	1.2222
Alpha-mixture ($\alpha = 0.1$)	0.6703	96.19	*0.1115	*82.93	*0.4142
Adaptive log-mixture	1.6459	84.08	0.5158	17.74	1.8370
Geometric ($\beta = 0.3$)	3.8978	*5.71	0.1938	1.76	0.8694
Fixed-mixture ($\lambda = 0.3$)	*0.0074	32.15	1.3819	2.05	4.6813
Fixed-mixture ($\lambda = 0.9$)	0.9630	82.28	0.1497	61.42	0.5597
Balanced (1)	7.1919	22.87	1.6222	0.94	5.8876
Balanced (3)	2.4692	57.17	1.7198	11.33	5.9600

Table 2: All proposal distributions and their calculated metrics, for $P = \text{Qwen2.5-1.5B}$, $Q = \text{Qwen2.5-0.5B}$ with temperature 0.4 prompted with the phrase “An interesting fact is that”. Experiments consist of $M = 1000$ samples drawn from the described distribution, where each sample is cut off after 20 tokens. Score combines accuracy (70%) and variance (30%).* indicates best performance in each metric, apart from the baseline. Bold font indicates second best performance.

which could be caused by the bias of importance sampling, or just due to very low rates of convergence.

No distribution performed as well as Naïve Monte Carlo sampling, which had the lowest estimated variance, as well as the second most accurate estimate. In fact, the two proposal distributions with the lowest amount of variance were those directly sampling from P with a 90% chance.

The only other particularly interesting proposal was the Fixed-Mixture ($\lambda = 0.3$) proposal, which resulted in a very low error compared to the baseline. On top of this, it had a very low weighted squared error. However, its low ESS score indicates very low weights, so the WSE might have been mostly affected by a few lucky samples.

5 Conclusion and Future Direction

While experiments on synthetic data showed a lot of promise for proposed importance sampling distributions, experiments on real LLMs did not reflect the same advantages. This illustrates the difficulty of analyzing the KL divergence between language models, due to their complexity.

Additionally, we believe that we made several mistakes which made it hard to find distributions that performed as well as naïve sampling.

1. **Divergence between P and Q :** Even though P and Q were from the same model family, $KL(P||Q)$ was significant, and samples could accumulate large variance very quickly. This meant that methods weighing towards large values of $|\log P/Q|$ often ended up overshooting by a long shot and ending up with tiny weights.
2. **Numerical Stability:** As mentioned earlier, our Qwen2.5 experiments were run using 16-bit floating point precision, for the sake of computational efficiency. While this is acceptable for applications of LLMs, this is just not enough information space to properly represent the LLM’s output logits, which were vectors over more than 10^5 possible tokens.

In hindsight, we believe we should have restricted P and Q to output a smaller set of tokens, enough to form simple sentences, but a small enough number to make our computations simpler.

5.1 Future Direction:

The most obvious and perhaps simplest solutions to our problems are two-fold:

1. **Use LMs P and Q with less divergence between them.** This would allow our calculations to converge more quickly, as well as alleviate numerical stability issues, since log difference values would stay smaller. For example, Amini et al. (Amini et al., 2025) used GPT-2.0 and a fine-tuned version of GPT-2.0, rather than two different model sizes.
2. **Use more computational power.** This includes running larger sample-sizes, running models with more numerical precision, and testing more potential distributions. This would increase our confidence in our results, as well as possibly provide better grounds for direct importance sampling, so we wouldn't have to use SNIS. Finally, with more computational power, we could combine Rao-Blackwellization with our proposal distributions, and still be able to compare their results against a baseline (naïve sampling with more samples).

While our results are inconclusive, we believe that more testing on new and old distributions could bring very interesting results.

References

- Sergios Agapiou, Omiros Papaspiliopoulos, Daniel Sanz-Alonso, and Andrew M Stuart. Importance sampling: Intrinsic dimension and computational cost. *Statistical Science*, pp. 405–431, 2017.
- Afra Amini, Tim Vieira, and Ryan Cotterell. Better estimation of the kullback–leibler divergence between language models. *arXiv preprint arXiv:2504.10637*, 2025. URL <https://arxiv.org/abs/2504.10637>.
- Nicolas Chopin, Omiros Papaspiliopoulos, et al. *An introduction to sequential Monte Carlo*, volume 4. Springer, 2020.
- Katsuma Inoue, Soh Ohara, Yasuo Kuniyoshi, and Kohei Nakajima. Transient chaos in bidirectional encoder representations from transformers. *Phys. Rev. Res.*, 4:013204, Mar 2022. doi: 10.1103/PhysRevResearch.4.013204. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.4.013204>.
- Nathan Kallus and Masatoshi Uehara. Intrinsically efficient, stable, and bounded off-policy evaluation for reinforcement learning. *Advances in neural information processing systems*, 32, 2019.
- Solomon Kullback and Richard A Leibler. On information and sufficiency. *The annals of mathematical statistics*, 22(1):79–86, 1951.

6 Appendix

6.1 Variance with correction factor

We note that while asymptotic variance is used to estimate the variance since we're computing finite sample variance, a correction factor of $\frac{ESS}{ESS-1}$ could be applied in line with the

typical sample variance formula:

$$\begin{aligned}\widehat{\text{Var}}[\hat{\mu}_{\text{SNIS}-\text{finite}}] &= \frac{1}{M} \frac{\frac{1}{M-1} \sum_i w_i^2 (f_i - \hat{\mu}_{\text{SNIS}})^2}{\frac{1}{M^2} (\sum_i w_i)^2} \\ &= \frac{M}{M-1} \sum_{i=1}^M w_i'^2 (f_i - \hat{\mu}_{\text{SNIS}})^2.\end{aligned}$$

The well performing estimators have high ESS so this correction has minimal effects on their variance estimates.

We note also that most estimators achieve lower variance with higher temperatures but with significantly lower estimates of KL divergence compared to the baseline estimate. The Qwen models often start producing nonsensical outputs or outputs in Mandarin, which is not desirable with an English prompt.

6.2 Bias calculation

Let $\hat{\mu}_{\text{SNIS}}$ be the self-normalized importance sampling estimate for $\mu = \mathbb{E}_{\mathbf{x} \sim p}[f(\mathbf{x})]$ using the sampling distribution r :

$$\hat{\mu}_{\text{SNIS}} := \frac{\sum_{i=1}^n w_i f_i}{\sum_{i=1}^n w_i} = \frac{\bar{W}_f}{\bar{W}}$$

where $\mathbf{x}_i \sim r$, and

$$\begin{aligned}f_i &:= f(\mathbf{x}_i), & w_i &:= P(\mathbf{x}_i)/r(\mathbf{x}_i), \\ \bar{W}_f &:= \frac{1}{n} \sum_i w_i f_i, & \bar{W} &:= \frac{1}{n} \sum_i w_i.\end{aligned}$$

We define the following notation:

$$\begin{aligned}\mu_w &:= \mathbb{E}[w] \text{ (mean of weights)} \\ \mu_{wf} &:= \mathbb{E}[wf] \text{ (mean of weighted function values)} \\ \delta_W &:= \bar{W} - \mu_w \text{ (deviation of sample mean from true mean)} \\ \delta_{Wf} &:= \bar{W}_f - \mu_{wf} \text{ (deviation of weighted sample mean)}\end{aligned}$$

Then we can write $\hat{\mu}_{\text{SNIS}}$ as

$$\hat{\mu}_{\text{SNIS}} = \frac{\mu_{wf} + \delta_{Wf}}{\mu_w + \delta_W}$$

For a function of two variables, $g(x, y)$, its Taylor expansion around a point (a, b) is

$$\begin{aligned}g(x, y) &\approx g(a, b) + g_x(a, b)(x - a) + g_y(a, b)(y - b) \\ &\quad + \frac{1}{2!} [g_{xx}(a, b)(x - a)^2 + 2g_{xy}(a, b)(x - a)(y - b) \\ &\quad + g_{yy}(a, b)(y - b)^2] + \dots\end{aligned}$$

Here,

$$g(\delta_W, \delta_{Wf}) = \frac{\mu_{wf} + \delta_{Wf}}{\mu_w + \delta_W}.$$

Taking the Taylor expansion around the point $(\delta_W = 0, \delta_{Wf} = 0) = (0, 0)$, we get

$$\begin{aligned}g(\delta_W, \delta_{Wf}) &\approx \frac{\mu_{wf}}{\mu_w} + \frac{1}{\mu_w} \delta_{Wf} - \frac{\mu_{wf}}{\mu_w^2} \delta_W \\ &\quad + \frac{\mu_{wf}}{\mu_w^3} \delta_W^2 - \frac{1}{\mu_w^2} \delta_W \delta_{Wf} + \dots\end{aligned}$$

Now take the expectation of the expansion:

$$\begin{aligned}\mathbb{E}[\hat{\mu}_{\text{SNIS}}] &\approx \frac{\mu_{wf}}{\mu_w} + \frac{1}{\mu_w} \mathbb{E}[\delta_{Wf}] - \frac{\mu_{wf}}{\mu_w^2} \mathbb{E}[\delta_W] \\ &\quad + \frac{\mu_{wf}}{\mu_w^3} \mathbb{E}[\delta_W^2] - \frac{1}{\mu_w^2} \mathbb{E}[\delta_W \delta_{Wf}]\end{aligned}$$

Notice that first-order terms vanish, since

$$\begin{aligned}\mathbb{E}[\delta_W] &= \mathbb{E}[\bar{W} - \mu_w] = 0 \\ \mathbb{E}[\delta_{Wf}] &= \mathbb{E}[\bar{W}_f - \mu_{wf}] = 0.\end{aligned}$$

We now compute second order moments:

$$\begin{aligned}\mathbb{E}[\delta_W^2] &= \text{Var}(\bar{W}) = \frac{1}{n} \text{Var}(w) = \frac{1}{n} (\mathbb{E}[w^2] - \mu_w^2) \\ \mathbb{E}[\delta_W \delta_{Wf}] &= \text{Cov}(\bar{W}, \bar{W}_f) = \frac{1}{n} \text{Cov}(w, wf) = \frac{1}{n} (\mathbb{E}[w^2 f] - \mu_w \mu_{wf}) \\ \mathbb{E}[\hat{\mu}_{\text{SNIS}}] &\approx \frac{\mu_{wf}}{\mu_w} + \frac{\mu_{wf}}{\mu_w^3} \cdot \frac{1}{n} (\mathbb{E}[w^2] - \mu_w^2) - \frac{1}{\mu_w^2} \cdot \frac{1}{n} (\mathbb{E}[w^2 f] - \mu_w \mu_{wf}) \\ &= \frac{\mu_{wf}}{\mu_w} + \frac{1}{n} \left[\frac{\mu_{wf} \mathbb{E}[w^2]}{\mu_w^3} - \frac{\mu_{wf} \mu_w^2}{\mu_w^3} - \frac{\mathbb{E}[w^2 f]}{\mu_w^2} + \frac{\mu_w \mu_{wf}}{\mu_w^2} \right] \\ &= \frac{\mu_{wf}}{\mu_w} + \frac{1}{n} \left[\frac{\mu_{wf} \mathbb{E}[w^2]}{\mu_w^3} - \frac{\mathbb{E}[w^2 f]}{\mu_w^2} \right].\end{aligned}$$

Note that $\mu = \mathbb{E}_P[f] = \frac{\mu_{wf}}{\mu_w}$, since $\mathbb{E}_r[wf] = \mathbb{E}_P[f]$ and $\mathbb{E}_r[w] = 1$. Thus we have that

$$\mathbb{E}[\hat{\mu}_{\text{SNIS}}] = \mu + \frac{1}{n} \left[\frac{\mu \mathbb{E}[w^2]}{\mu_w^2} - \frac{\mathbb{E}[w^2 f]}{\mu_w^2} \right] + O\left(\frac{1}{n^2}\right)$$

And since $\mu_w = \mathbb{E}[w] = 1$ for properly normalized importance sampling, we get that the bias is proportional to $\frac{1}{n}$:

$$\begin{aligned}\mathbb{E}[\hat{\mu}_{\text{SNIS}}] &= \mu + \frac{1}{n} [\mu \mathbb{E}[w^2] - \mathbb{E}[w^2 f]] + O\left(\frac{1}{n^2}\right) \\ \text{Bias} &= \frac{1}{n} \left[\frac{\mathbb{E}[w^2 f]}{\mathbb{E}[w]^2} - \frac{\mathbb{E}[wf] \mathbb{E}[w^2]}{\mathbb{E}[w]^3} \right] + O\left(\frac{1}{n^2}\right)\end{aligned}$$